

- 3 (short) An op-amp circuit is connected with voltage source v_{in} through a co-axial cable in Fig. 3. The source has an output resistance R_s of $500\ \Omega$. Determine the midband voltage gain v_{in}/v_{out} and the 3 dB frequency of the circuit. State any assumptions made. The op-amp is ideal. [10]

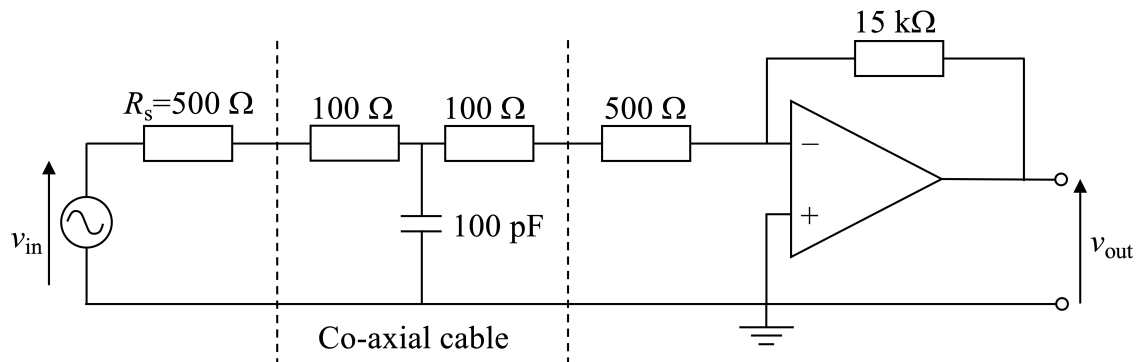


Fig. 3